

EVAN P. GRAY

GRAPHICS PROGRAMMER

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SKILLS

Languages:	C++, C, C#, GLSL HLSL, Slang
Graphics APIs:	Vulkan, Metal, OpenGL
Tools:	NVIDIA Nsight Graphics, RenderDoc, Xcode Metal Debugger
General:	Debugging, Optimization, Organization, Collaboration

WORK EXPERIENCE

Graphics Programmer

May 2026 - Present

The Forge Interactive, Inc.

- Independently ported a legacy production codebase to MacOS and Android mobile devices, utilizing Metal and Vulkan respectively.
- Leveraged Renderdoc and Xcode's Metal Debugger to resolve critical rendering issues.
- Resolved severe GPU synchronization issues in the rendering framework's Steam Deck build.

PROJECTS

Sole Graphics Programmer

Solo Project

RayTracer - 3D Hardware Accelerated Raytraced Lighting Engine

- Implemented modern hardware accelerated raytracing through the Vulkan API's acceleration structures to render high-fidelity 3D models in real time.
- Developed features like shadow rays, transparency, recursive reflections, and normal mapping.
- Integrated a Physically Based Rendering lighting model to produce realistic surface materials.
- Utilized NVIDIA Nsight Graphics' toolkit to debug and optimize raytracing code.

Graphics Programmer, Assistant Producer

9 Member Team

NiteLite - 3D Custom Game Engine in C++

- Collaborated with another developer to implement real-time 3D object rendering using the Vulkan API that can handle 1000's of complex meshes above 200fps.
- Researched and implemented terrain creation using tessellated heightmaps to allow designers to handcraft engaging maps that are deformable during runtime.
- Utilized RenderDoc to diagnose issues and ensure future stability within rendering code as well as identifying and fixing performance bottlenecks.
- Led our team in developing a user story of the final product, ensuring a cohesive understanding of project outcomes, increasing team motivation, and identifying more than 10 unplanned tasks.



Sole Graphics Programmer, Producer

7 Member Team

KnightLight - 2D Puzzle Platformer in a Custom C++ Game Engine

- Prototyped and created a custom 2D lighting algorithm in OpenGL enabling designers to implement light-based mechanics with support for dozens of dynamic, customizable and real-time lights.
- Increased frame rate by 1000% by implementing compute shaders into the lighting system, allowing for hundreds of thousands of rays per frame.
- Led a cross-functional team of 7 members and resolved workflow issues throughout the project.

EDUCATION

**Bachelor of Science in Computer Science
in Real-Time Interactive Simulation**

DigiPen Institute of Technology

04/2026
GPA: 3.99